

QUESTION 1

Preparing a Greenhouse Gas Inventory

What to count, where to draw the line, and what the answer changes.

Worked on a 120,000 barrel-per-day Nigerian refinery

GROUP C (29 MEMBERS)

Three scopes. One fence.

The scopes exist so that two companies never count the same tonne twice.

SCOPE 1

Direct

Everything burned, flared, vented or leaked inside your fence.

SCOPE 2

Indirect — energy

The power station that runs your lights.

SCOPE 3

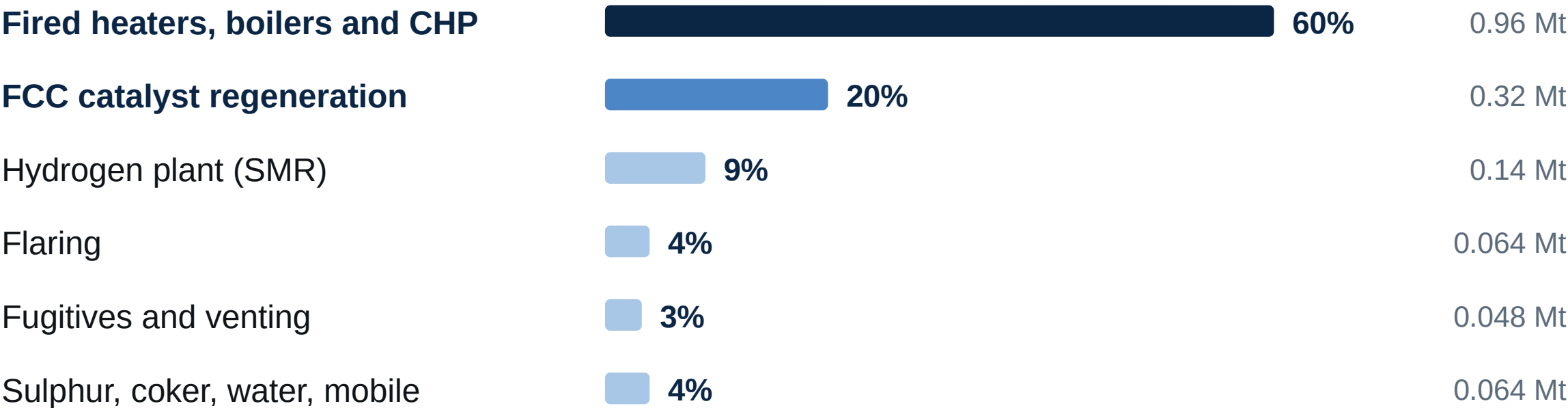
Indirect — everything else

Your crude supplier, and every vehicle burning the diesel you sold.

Scope 1 and 2 are what you control. Scope 3 is what you cause.

Two sources carry four fifths of it.

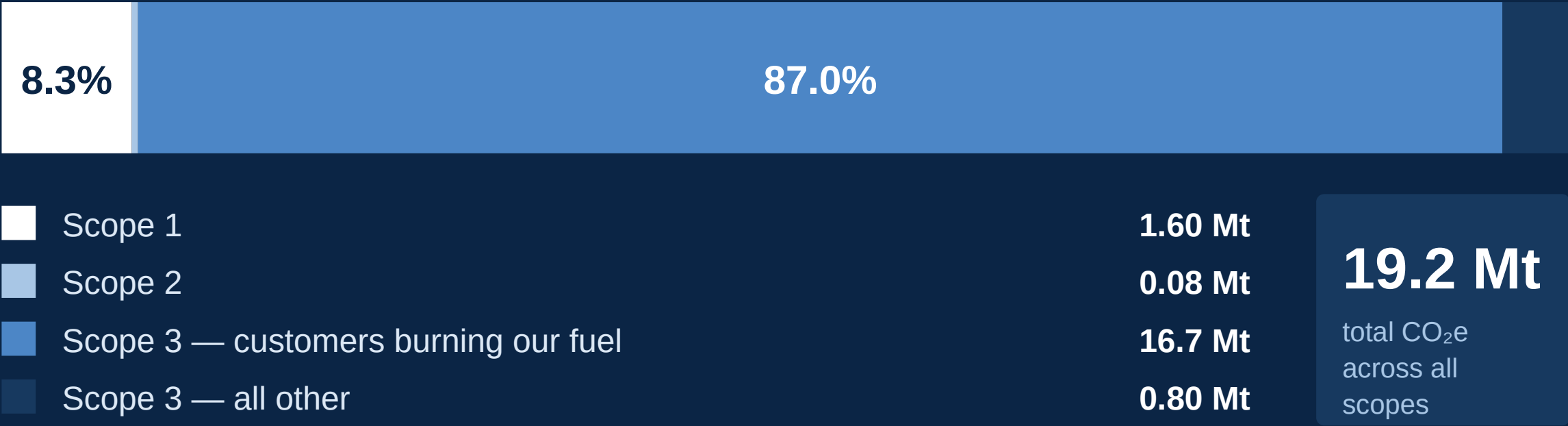
A refinery is not one chimney. It is about a dozen distinct mechanisms.



The two most often missed: the FCC burns no purchased fuel, and the hydrogen plant releases CO₂ from the reaction, not the flame.

THE NUMBER THAT CHANGES THE CONVERSATION

87% of the footprint leaves the gate in a tanker.



Every efficiency lever works on the 8.8%. Moving the 87% is a question about what we sell.

Measure it well, then act on it.

An inventory that changes no decision was a bookkeeping exercise.

HOW WE MEASURE

- 1 Continuous monitoring in the stack
- 2 Mass balance — carbon in, carbon out
- 3 Engineering calculation
- 4 Published emission factor

Best evidence at the top. Say which you used.

WHAT WE DO ABOUT IT

Energy and heat integration	3–8%
Methane leak detection	1–3%
Flare gas recovery	2–4%
Capture on the hydrogen plant	7–9%

Cut to Scope 1, cheapest first.

An inventory is a strategy document that happens to be written in tonnes.